

KAITIAN CHAO

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PROFESSIONAL SUMMARY

Kaitian Chao is a second-year Robotics master's student at the University of Pennsylvania's GRASP Laboratory, focusing on **robot learning, 3D perception, and control**. His research aims to build generalist robot agents that can reason in language and code, orchestrate modular robotics tools, and continually learn new concepts and skills from experience and sparse human feedback. He has extensive experience in deep learning and robotics research, with projects spanning generalist manipulation with VLM/VLA-driven policies, drone racing with reinforcement learning, F1tenth autonomous driving, robotic fish modeling and control, diffusion models, and large language models, all aimed at advancing AI-driven capabilities. His passion lies in leveraging cutting-edge AI and vision technologies to develop super-intelligent robotics and software that can significantly enhance productivity and help people in real life.

EDUCATION

University of Pennsylvania

August 2024 - May 2026

Master's, Robotics

- Machine Perception
- Control and Optimization with Applications in Robotics
- F1/10 Autonomous Racing Cars
- Learning in Robotics
- Advanced Robotics
- Physical Intelligence: Science and Systems (Reinforcement Learning)

ShanghaiTech University

September 2020 - July 2024

Bachelor's, Electrical Engineering

GPA: 3.73

- Stochastic Processes
- Introduction to Communication Systems

University of California - Berkeley

August 2022 - May 2023

Certification, Electrical Engineering and Computer Science

GPA: 3.96

- Designing, Visualizing and Understanding Deep Neural Networks
- Feedback Control Systems
- Optimization Models in Engineering
- Data Structures

PUBLICATIONS

MAESTRO: Orchestrating Robotics Modules with Vision-Language Models for Zero-Shot Generalist Robots

Junyao Shi*, Rujia Yang*, **Kaitian Chao***, Bingqing Wan, Yifei Shao, Jiahui Lei, Jianing Qian, Long Le, Pratik Chaudhari, Kostas Daniilidis, Chuan Wen, Dinesh Jayaraman

IEEE International Conference on Robotics and Automation (ICRA), 2026 (Under review)

CoRL RoboArena Workshop, 2025 (Oral)

NeurIPS SpaVLE workshop, 2025 (Oral)

Learning Flow-Adaptive Dynamic Model for Robotic Fish Swimming in Unknown Background Flow

Kaitian Chao*, Xiaozhu Lin*, Xiaopei Liu, Yang Wang

IEEE International Conference on Intelligent Robots and Systems (IROS), 2025

Ambient Flow Perception of Freely Swimming Robotic Fish Using an Artificial Lateral Line System

Hongru Dai*, Xiaozhu Lin*, **Kaitian Chao**, Yang Wang

IEEE International Conference on Robotics and Automation (ICRA), 2025

RESEARCH & ENGINEERING PROJECTS

Maestro: Orchestrating Robotics Modules with Vision-Language Models for Zero-Shot Generalist Robots **Philadelphia, PA, USA**

Graduate Researcher at GRASP PAL Lab, advised by Prof. Dinesh Jayaraman

May 2025 - Present

- Developed Maestro, a VLM-driven modular generalist policy that treats mature perception, planning, and control systems as tools and uses a coding VLM to orchestrate them into closed-loop policies for diverse tabletop and mobile manipulation tasks, matching or surpassing leading VLA models in zero-shot settings without any robot teleoperation data.
- Built an extensible robotics toolbox and agent interface that integrates new perception/planning/control modules and VLAs with minimal engineering, enabling code-level adaptation from limited real-world trials and laying the foundation for large-scale automatic data collection and policy distillation for future general-purpose robots.

Learning Flow-Adaptive Dynamic Model for Robotic Fish Swimming in Unknown Background Flow **Shanghai, China**

Undergraduate Researcher at MAgIC Lab

October 2023 - October 2024

- Developed a general and effective method for the dynamic modeling of robotic fish in the background flow environment, utilizing the Domain Adversarial Meta Learning algorithm to achieve flow-agnostic and adaptive modeling.
- Designed and trained deep reinforcement learning policy to enable robust control of robotic fish swimming with different flow conditions.
- First author of a paper on learning-based dynamic modeling of robotic fish in flow environment, accepted by the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)

Autonomous Racing with MPPI and Hybrid Control – F1TENTH Race Car **Philadelphia, PA, USA**

February 2025 - May 2025

- Led the development and deployment of a full-stack autonomous driving system on a 1/10-scale F1TENTH car, integrating LiDAR/vision-based perception, SLAM, particle filtering for localization, and RRT* for obstacle-avoidance planning.
- Implemented MPPI and a hybrid Pure Pursuit–MPC controller for real-time collision-free trajectory optimization, complemented by reinforcement and imitation learning to refine racing strategies.
- Optimized hardware dynamics (suspension, tires) and achieved robust, low-latency performance, securing a top-tier finish at the 24th Roboracer Grand Prix, ICRA 2025.

Vision-Based Motion Planning for Agile Drone Navigation in Cluttered Environments **Philadelphia, PA, USA**

March 2025 - May 2025

- Developed a complete vision-based navigation pipeline for quadrotors by integrating onboard mapping, A* path planning, and differential flatness–based geometric control to execute collision-free, time-optimal trajectories.

Dynamic Pick-and-Place Motion Planning with 7-DOF Franka Emika Panda robot arm **Philadelphia, PA, USA**

November 2024 - December 2024

- Developed an end-to-end pipeline for real-time 7-DOF robot to pick and place static and dynamic blocks using ROS and Gazebo framework. Implemented motion planning using Geometric Inverse Kinematic and bi-directional RRT for tight and fast control.

Ambient Flow Perception of Freely Swimming Robotic Fish Using an Artificial Lateral Line System **Shanghai, China**

Undergraduate Researcher at MAgIC Lab

October 2023 - September 2024

- Developed the first ALLS-based ambient flow classifier for robotic fish, training an LSTM on only 3.5 minutes of swimming sensor data from 5 pressure sensors to achieve 81.25% accuracy across 8 flow categories (0.08–0.18 m/s), establishing a robust and generalizable perception framework. Research accepted for publication at IEEE ICRA 2025

PROFESSIONAL EXPERIENCE

Penn Aerial Robotics

Philadelphia, PA, USA

Software Team Lead

October 2024 - Present

- Lead the software team to design autonomous control, perception and machine learning algorithm for the UAV plane.
- Participated in the SAE Aero Design West 2025 competition, achieving 2nd place nationally with the team; successfully completed 3 takeoffs and 1 full flight sequence, lifting over 110 lbs in total payload.
- Organized in-house competition of drone racing: program drones to fly through hoop courses using advanced computer vision, path planning and control algorithm.

SKILLS

Skills: Python, Pytorch, ROS2, IsaacSim, IsaacLab, Gazebo, Pybullet, DROID, Java, C/C++, MATLAB, C#, Unity, Git, Machine learning, Deep reinforcement learning, 3D computer vision, SLAM, NERF, Excel/Numbers/Sheets

Languages: Chinese, English